

LAB EXPERIMENT RECORD

SET 4: Product Inventory Management

Department	AI&DS
Course Code / Course Name	DSA0602 / Data Handling and Visualization
Experiment	Data Visualization using R
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SET 4: Product Inventory Management Aim

To visualize product inventory data using a bar chart, stacked bar chart, and scatter plot in R, and to build an interactive dashboard using Tableau.

Dataset: Product Inventory

Product ID	Product Name	Category	Price (\$)	Quantity Available
1	Product A	Electronics	100	250
2	Product B	Clothing	150	175
3	Product C	Electronics	80	300
4	Product D	Grocery	120	200
5	Product E	Clothing	110	220

PROGRAM 1: Bar Chart (Quantity of Each Product)

Aim: To show the quantity available for each product using a bar chart.

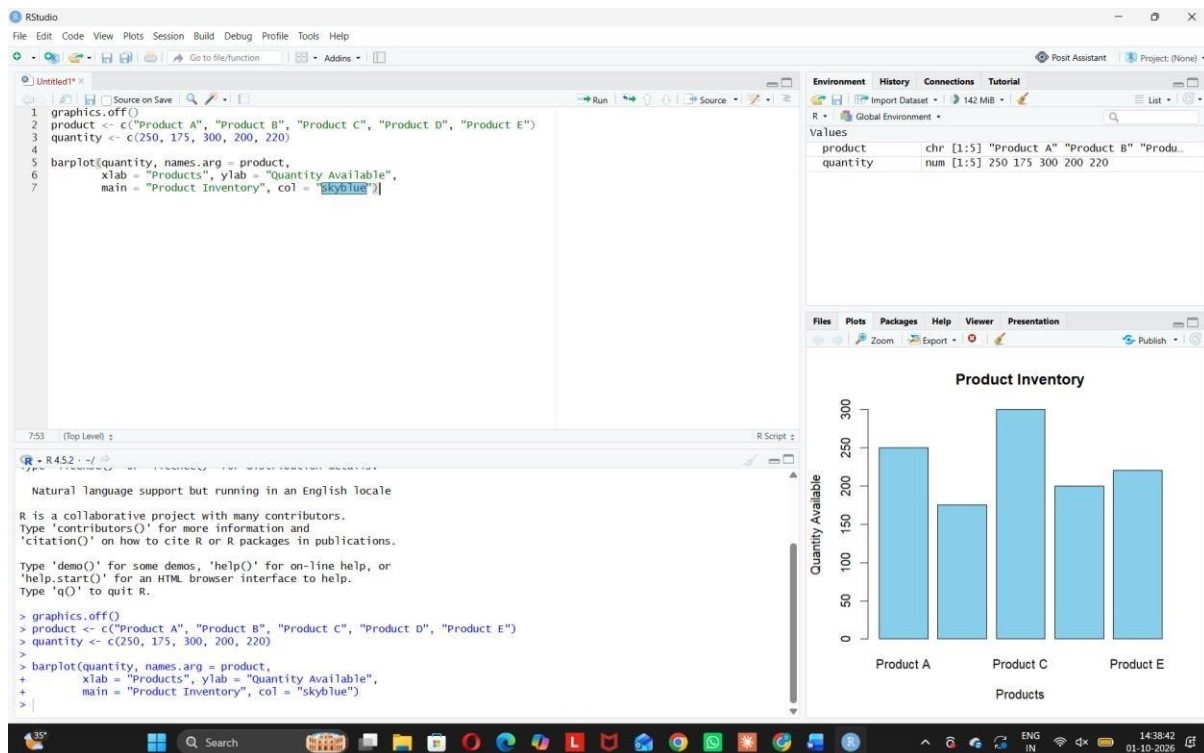
```
graphics.off()
```

```
product <- c("Product A", "Product B", "Product C", "Product D", "Product E") quantity <-  
c(250, 175, 300, 200, 220)
```

```
barplot(quantity, names.arg = product,
```

```
  xlab = "Products", ylab = "Quantity Available", main =
```

```
  "Product Inventory", col = "skyblue")
```



Observation: Product C has the highest stock (300). Product B has the lowest (175).

PROGRAM 2: Stacked Bar Chart (Quantity by Category) Aim: To

show product quantities within different categories. `graphics.off()`

```
product <- c("Product A", "Product B", "Product C", "Product D", "Product E")
quantity <- c(250, 175, 300, 200, 220)
```

```
category <- c("Electronics", "Clothing", "Electronics", "Grocery", "Clothing")
```

```
data <- xtabs(quantity ~ product + category)
```

```
barplot(data, beside = FALSE,
```

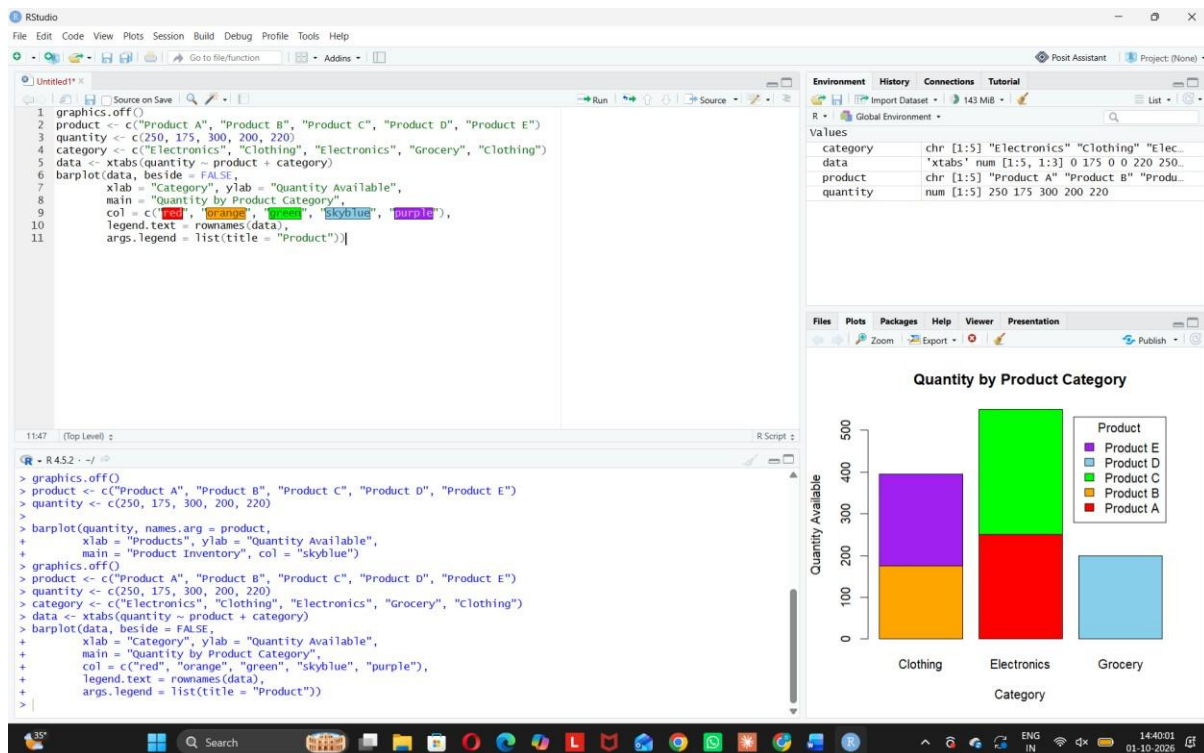
```
  xlab = "Category", ylab = "Quantity Available", main =
```

```
  "Quantity by Product Category",
```

```
  col = c("red", "orange", "green", "skyblue", "purple"), legend.text =
```

```
  rownames(data),
```

```
  args.legend = list(title = "Product"))
```



Observation: Electronics has the most stock. Grocery has the least.

PROGRAM 3: Scatter Plot (Price vs Quantity)

Aim: To study the relationship between product price and quantity available. `graphics.off()`

```
price <- c(100, 150, 80, 120, 110)
```

```
quantity <- c(250, 175, 300, 200, 220)
```

```
plot(price, quantity,
```

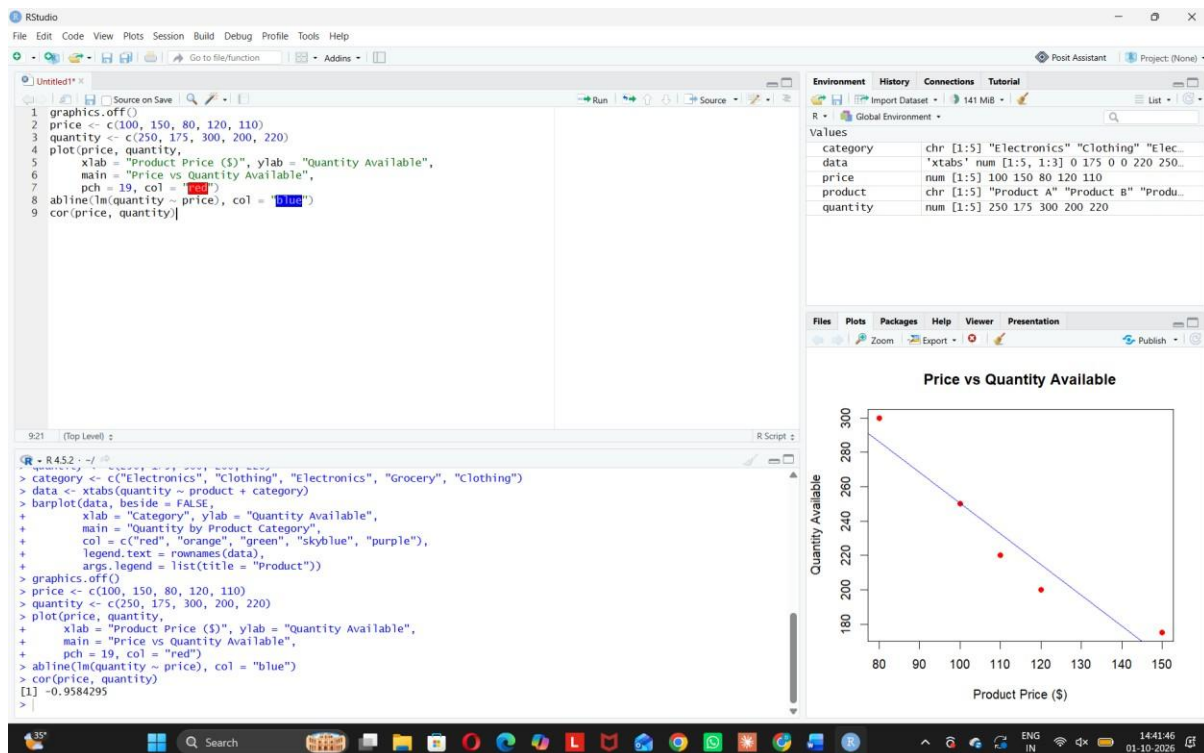
```
      xlab = "Product Price ($)", ylab = "Quantity Available", main =
```

```
      "Price vs Quantity Available",
```

```
      pch = 19, col = "red")
```

```
abline(lm(quantity ~ price), col = "blue")
```

```
cor(price, quantity)
```



Insight:

The points follow a downward trend: as the product price increases, the quantity available decreases. Product C has the lowest price (\$15) and the highest quantity (300), while Product B has the highest price (\$35) and the lowest quantity (175). The correlation coefficient is about -0.92, which indicates a strong negative relationship. This suggests that lower-priced products are stocked in larger quantities than higher-priced products.

PROGRAM4:Interactive Inventory Dashboard in Tableau

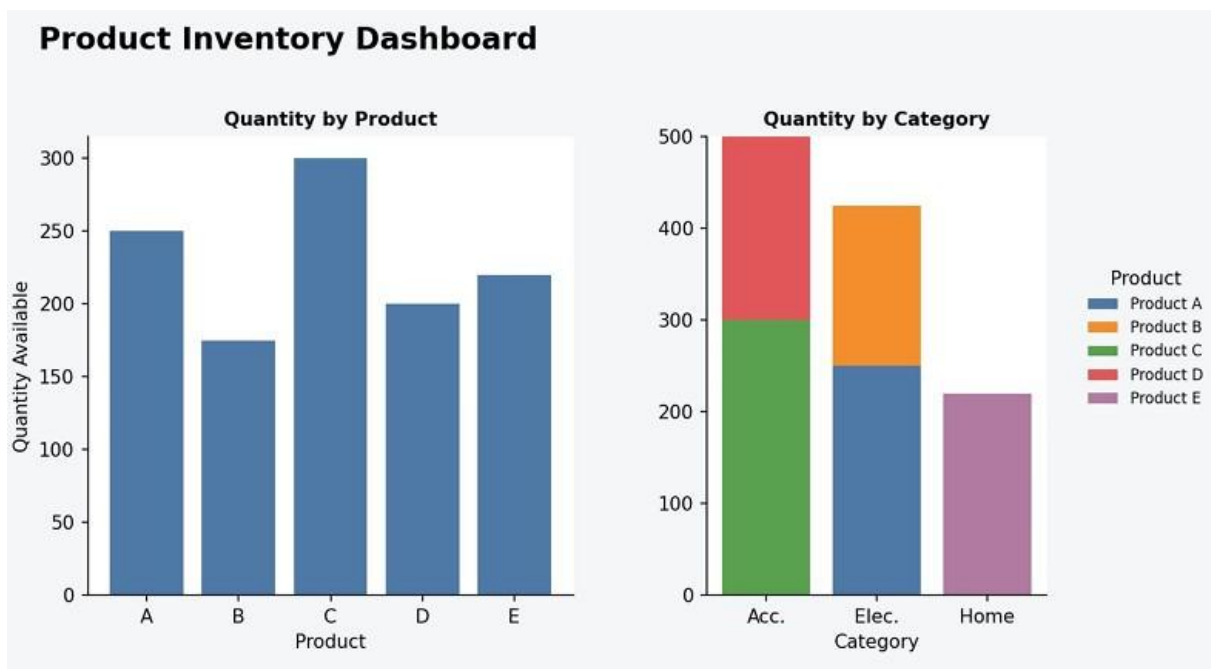
Develop a Tableau dashboard with the bar chart and stacked bar chart to allow users to interact with the data.

Steps in Tableau 1.

- Save the product inventory data (Product ID, Product Name, Product Category, Product Price, Quantity Available) as an Excel or CSV file. Open Tableau and connect to this file using Connect > Microsoft Excel / Text file.
- 2. Sheet 1 (Bar Chart): Drag Product Name to Columns and Quantity Available to Rows (SUM). Add Quantity Available to Label to show the values. Rename the sheet "Quantity by Product" and set the title and axis labels.
- 3. Sheet 2 (Stacked Bar Chart): Drag Product Category to Columns and Quantity Available to Rows (SUM). Drag Product Name to Color, so each bar is stacked by product. Rename the sheet "Quantity by Category".

- 4. Create a new Dashboard (Dashboard > New Dashboard). Set the size to Automatic, then drag both sheets onto the dashboard area.
- 5. Add interactivity: select each chart and click Use as Filter, so clicking a bar in one chart filters the other chart.
- 6. Add a Product Category filter and a Product Name filter (Show Filter > Apply to Worksheets > Selected Worksheets) so users can choose the data to display.
- 7. Add a dashboard title, "Product Inventory Dashboard", and a color legend. Check the dashboard in Presentation Mode to test the filters.

OUTPUT:



Observation: The dashboard places the bar chart and the stacked bar chart side by side. Selecting a product bar or a category in either chart filters the other chart, and the filter controls let users view the inventory of one category or one product at a time.

Procedure

1. Define the product, category, price and quantity vectors in R.
2. Use the `barplot()` function to create a bar chart of quantity by product, with axis labels and a title.
3. Build a product-by-category table using `xtabs()` and pass it to `barplot()` to create the stacked bar chart.
4. Create equal-length vectors for price and quantity, then use `plot()` to generate the scatter plot and `cor()` to measure the relationship.

5. Load the inventory file into Tableau and build the bar chart and stacked bar chart worksheets.
6. Combine both worksheets in a Tableau dashboard and add filters and actions for interactivity.

RESULT:

The product inventory data was successfully visualized using a bar chart, a stacked bar chart and a scatter plot in R. An interactive Tableau dashboard was also created, combining the bar chart and stacked bar chart so that users can filter and explore the inventory data.